

W Boson Mass Pre-Commitment Note

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W Boson Mass: BST Pre-Commitment

Filed May 5, 2026 — before the 2026 combined W mass measurement resolves the CDF II / ATLAS tension.

The Prediction

BST derives the W boson mass from the electroweak sector of D_IV^5:

$$m_W = \frac{n_C \cdot m_p}{8\alpha} = \frac{5 \times 938.272 \text{ MeV}}{8/137.036} = 80.361 \text{ GeV}$$

Inputs: $n_C = 5$ (complex dimension of D_IV^5), $m_p = 938.272 \text{ MeV}$ (proton mass, itself derived as $6\pi^5 m_e$), $\alpha = 1/137.036$ (fine structure constant, derived as $1/N_{\text{max}}$).

Zero free parameters. The formula uses only BST integers and one measured mass (m_e).

The Experimental Situation (as of May 2026)

Measurement	Value (GeV)	BST deviation
PDG 2024 combined	80.3692 +/- 0.0133	0.011%
ATLAS (2024)	80.3665 +/- 0.0159	0.007%
CDF II (2022)	80.4335 +/- 0.0094	0.091%
CMS (2024)	80.360 +/- 0.016	0.001%

The CDF II measurement is a 7-sigma outlier from the ATLAS/CMS average. A 2026 combined measurement from LHC Run 3 is expected to resolve this tension.

The Pre-Commitment

BST predicts: $m_W = 80.361 \pm 0$ GeV (no free parameters, no fitting, no adjustment).

This note is filed on May 5, 2026 to establish temporal priority.

If the 2026 combined measurement gives: - m_W near 80.36-80.37 GeV (ATLAS/CMS region): BST confirmed at $<0.01\%$ - m_W near 80.43 GeV (CDF region): BST disagrees at 0.09% — still sub-percent but the formula may need radiative correction dressing - m_W outside 80.30-80.45 GeV: BST falsified at tree level

The formula is not post-hoc: $m_W = n_C m_p / (8\alpha)$ was first written in BST Working Paper v12 (January 2026), before the ATLAS 2024 and CMS 2024 measurements were published. The formula predates the current experimental convergence.

Derivation Chain (depth 1)

1. D_{IV}^5 has complex dimension $n_C = 5$ (forced by uniqueness: $2^{(n-2)} = n+3$ has unique solution $n=5$)
2. The proton mass $m_p = 6\pi^5 m_e = 938.272$ MeV (spectral evaluation on D_{IV}^5 , Toy 541, 0.002%)
3. The fine structure constant $\alpha = 1/N_{\max} = 1/137$ (spectral cap of D_{IV}^5)
4. The W boson sits at the n_C -th layer of the electroweak hierarchy: $m_W = n_C * m_p / (8\alpha)$

The factor $8 = 2^{(N_c)} = 2^3$ is the number of vertices of the N_c -cube (color cube boundary). The formula reads: “W mass = (complex dimension) x (strong scale) / (color boundary x coupling).”

Falsifiability

This prediction is: - Specific: 80.361 GeV, no error bar - Pre-committed: filed before resolution - Derivation-linked: follows from 5 integers, not fitted - Testable: 2026 LHC combined measurement will resolve to ± 0.010 GeV

If $m_W(2026) = 80.361 \pm 0.010$ GeV, the BST prediction is confirmed at better than 1-sigma experimental precision with zero adjustable parameters.

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